Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/27/2007
PI Name:	Purdy, Ralph Ph.D.		
Project Title:	Effect of Microgravity on Vascular Signal Transduction		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
PI Email:	repurdy@uci.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	949-824-7653
Organization Name:	University of California - Irvine		
PI Address 1:	Department of Pharmacology, College of Medicine		
PI Address 2:	350C Med Surge II, zot 4625		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697-4625	**Congressional District:**	48
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-04
Start Date:	06/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NCE through 9/30/2007, per A. Chu (8/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CK29G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Hindlimb unweighting in the rat was used to simulate microgravity and arteries were isolated to study the effects of HU on vessel function, morphology and compliance. Evidence was obtained that HU: 1) decreases the expression of myosin heavy chain (MHC), myosin light chain 20 (MLC-20) and alpha actin protein expression and that this could account for the HU-induced vasoconstrictor hyporesponsiveness of rat abdominal aorta; 2) decreases MLC-20 protein expression and vessel wall compliance, but does not change morphometric parameters in the carotid artery; 3) decreases both alpha actin and MLC-20 protein expression, increases vessel wall compliance, but does not change morphometric parameters in the femoral artery; 4) causes a marked up-regulation of the contribution of nitric oxide mechanisms to the vasoconstrictor response of thoracic aorta; 5) has no effect on alpha ARs or alpha 2ARs in any vessel studied.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research Impact. It was found that microgravity exposure decreases, or has no effect on endothelial funciton in, and distal to the abdominal aorta. In contrast, endothelial function is increased and contributes to the vascular hyporesponsiveness caused by microgravity exposure in, and proximal to the thoracic aorta. This heterogeneous response to microgravity strongly argures angainst using therapies that target endothelium in order to treat/prevent microgravity-induced orthostatic intolerance. It was found that microgravity decreases the expression of one or more of the vascular contractile proteins, actin, myosin heavy chain or myosin light chain. In turn, evidence was obtained that this changed protein expression could underlie the vascular hyporesponsiveness associated with microgravity exposure. This strongly suggests that therapies that increase vascular contractile protein exprission, such as the PPAR gamma agonists, may be used to treat/prevent orthostatic intolerance. Earth Impacts. Orthostatic intolerance occurs in all patients who must undergo long term bed rest for recovery. Moreover, the animal model used throughout this study, the hindlimb unweighted rat, closely mimics the effects of bed rest. The present results strongly suggest that therapies that increase vascular contractile protein expression mlay be useful in these patients.
Task Progress:	The overall goal of the research was to understand the impact of microgravity on vascular function. Microgravity was simulated using hindlimb unweighting HLU in the rat. In one study, we looked at the effect of 20-day HLU on the carotid and femoral arteries. It was hypothesized that HLU would decrease contractile capacity by reducing the expression of contractile proteins and by changing the elastic properties of the carotid and femoral arteries. Thus, experiments were carried out in this study to measure the effects of HLU on alpha-actin, myosin light chain 20 (MLC-20), and myosin heavy chain (MHC) in the carotid and femoral arteries. HLU treatment did not significantly change the levels of alpha actin in either artery. However, HLU caused a marked decrease in myosin light chain 20 in both the carotid, and femoral arteries. Myosin heavy chain expression was unchanged by HLU in the carotid artery but was significantly decreased by HLU in the femoral artery. Experiments were carried out to determine length force relationships in carotid and femoral arteries. First, vessel rings 1 mm in length were placed under near-zero force in order to reshape them to two parallel sheets of tissue. The forces required to achieve this unstretched length, Lzero, were measured. HLU caused a nearly three-fold increase in the force required to reach Lzero in the carotid artery, indicating an increase in vessel stiffness. In contrast, HLU reduced the force required to reach Lzero in the femoral artery to less that half that in the control, indicating a decrease in vessel stiffness. The length-passive force relationship was also determined. The HLU rings exhibited significantly higher forces than control rings at all suprathreshold lengths. Thus, both the force required to achieve Lzero and the length-passive force measurements indicated that HLU increased carotid artery stiffness. The length-passive force relationships in the femoral artery were also determined. While passive force increased with increasing length, HLU had no effect, compared to control. Artery rings were placed at Lzero lengthened incrementally and exposed to 100 mM K+ at each increment to determine the length-active force relationship. In carotid artery, there was no contraction to 100 mM K+ at Lzero. However, the magnitude of force development increased with increasing length up to 0.6 mm and remained at a plateau thereafter. Moreover, there were no significant differences between control and HLU vessels. In contrast to the carotid artery, the femoral artery exhibited a contraction to 100 mM K+ at Lzero. This vessel also exhibited increases in active force with increasing length to a peak at 0.6 mm. Thereafter, there was a decrement in force with each further increase in length. The control and HLU force curves mirrored each other on the X axis. On the other hand, the magnitude of force on the Y axis was significantly less in HLU compared to control at all lengths. The unstretched length was determined by cutting both fresh and formalin-fixed rings open and measuring the length of the ring, representing the circumference. Diameters were determined by dividing the circumference by pi. Neither zero calcium Krebs's solution nor 100 μM sodium nitroprusside influenced the length. There were no significant differences between control and HLU carotid or femoral circumferential lengths in fresh tissue. Similar results were obtained in the formalin-fixed samples with the exception that the HLU femoral artery circumference was 4% less than that of the control femoral artery. It is also of interest that the fresh and formalin-fixed tissues had nearly identical circumferential lengths. The tissues fixed in formalin at maximum stretch returned to their original unstretched lengths during the two month storage in formalin. Medial thickness and medial cross-sectional area were measured in both control and HLU carotid and femoral arteries that had been isolated, pressurized to 100mm of Hg, formalin-fixed and prepared for histological analysis. HLU had no significant effect on these parameters in either vessel. Shrinkage and other fixation and treatment artifacts may have occurred, decreasing the reliability of measurements of vessel circumference. This parameter was measured in separate experiments (see above). On the other hand, the measurements of wall thickness and medial cross-sectional area were considered more reliable because of the identical treatment to which all vessels were subjected, and because of the noncompressible nature of the vessel wall. Thus, we conclude that HLU had no effect on either medial thickness or cross-sectional area. The total DNA content of the carotid and femoral arteries in both Control and HU-treated rats were determined. HLU has no effect on DNA. This suggests that cell number did not change with HLU treatment. Taken together, these data suggest that the HLU-induced decreases in both MLC-20 and MHC in femoral artery are responsible for the decreased contraction to 100mM K+ in HLU-treated femoral artery rings. In the carotid artery, the HLU-induced decrease in vessel wall compliance may counter any decrease in contractility caused by the decreased MLC-20 levels. In the study of the effects of HLU on the thoracic aorta, it was hypothesized that HLU would have no effect because HLU did not cause a contractile deficit. To our surprise, HLU increased both endothelium-mediated relaxation and vascular smooth muscle strength of contraction in Wistar thoracic aorta. These two effects cancelled each other out, accounting for the absence of a contractile effect. In contrast, in Sprague-dawley thoracic aorta, HLU increased

endothelium-mediated relaxation but had no smooth muscle effect. This provides an explanation for the well known, contractile deficit reported in the Thoracic aorta of Sprague-Dawley rats. The following results illustrate these findings.

CRCs for phenylephrine (PHE) in the Wistar rat revealed that HLU had no significant effect on contraction in the endothelium-intact thoracic aorta. In contrast, the maximal contraction of the endothelium-denuded HLU aorta was markedly greater than control. Similar results were obtained using norepinephrine (NE) as the agonist.

CRCs for the relaxing effects of acetylcholine were obtained in phenylephrine-precontracted, endothelium-intact arteries. Acetylcholine caused the expected concentration-dependent relaxation. However, HLU treatment had no effect. These experiments were repeated using a different muscarinic agonist, methylfurmethide (MFM), which is not a substrate for acetylcholine esterase and has no effect on nicotinic receptors. MFM caused a concentration-dependent relaxation. The relaxation to MFM, 0.003 to 0.1 µM, was significantly greater in HLU compared to control artery rings. Sodium nitroprusside (SNP), 0.03 to 0.3 µM, also caused a greater relaxation in HLU compared to control artery rings.

The Wistar strain of rats was used in the PI's study while previous work in other labs used the Sprague-Dawley strain. For comparison, experiments were repeated using thoracic aorta from Sprague-Dawley rats. The contraction of endothelium-intact thoracic aorta to NE was significantly less in HLU compared to control thoracic aorta. This was similar to the response of endothelium intact tissues to phenylephrine, although less in magnitude. Endothelium removal abolished the significant difference between HLU and control contractile responses to both PHE and NE. It is concluded that HLU increased smooth muscle contractile, and endothelium-dependent relaxing capacities equally in Wistar aorta, and had no effect on smooth muscle but increased endothelium-dependent relaxation in Sprague-Dawley aorta.

Bibliography Type:	Description: (Last Updated: 12/28/2007)
Articles in Peer-reviewed Journals	Hwang S, Shelkovnikov SA, Purdy RE. "Simulated microgravity effects on the rat carotid and femoral arteries: role of contractile protein expression and mechanical properties of the vessel wall." J Appl Physiol. 2007 Apr;102(4):1595-603. PMID: 17218426 , Apr-2007
Articles in Peer-reviewed Journals	Summers SM, Ulloa AS, Purdy RE. "Endothelial role in the thoracic aorta response to hindlimb unloading in rats." Aviat Space Environ Med. 2007 Dec;78(12):1103-7. PMID: 18064913 , Dec-2007
Articles in Peer-reviewed Journals	Shelkovnikov SA, Summers SM, Purdy RE. "Effect of Simulated Microgravity on Vascular Alpha Adrenoceptors." In revision for submission to J Pharmacol Exper Ther, January 2008. , Jan-2008
Articles in Peer-reviewed Journals	Hwang S, Kahwaji C, Sun A, Haddad F, Ulloa A, Goraya N, Huynh J, Le S, Nguyen T, Willis V, Purdy RE. "Simulated Microgravity Decreases Contractile Proteins and Increases Heavy Caldesmon in Rat Abdominal Aorta." In revision for submission to J Appl Physiol, January 2008. , Jan-2008
Books/Book Chapters	Meck JV, Purdy RE. "Endothelium in space." in "Endothelial biomedicine." Ed. W.C. Aird. Cambridge ; New York : Cambridge University Press, 2007. pp. 520-526, Jun-2007